CAPSTONE 46

Data Storytelling with Open Government Data

Case study: School Education in India

Project question: How can AI help turn public data into a clear, honest story that ordinary readers understand?

Target audience: General public, students, parents, teachers and local-community readers.

Complete project pack: research summary, verified indicators, four charts, data story, social media series, six-slide presentation, quiz, automation script, prompt library, and AI verification report.

Important scope note: National figures below are taken from the official UDISE+ 2023–24 national highlights. State/district CSVs were not downloaded and independently validated for this pack; no state-level values are fabricated.

1. Abstract

This project demonstrates how AI can help turn official school education statistics into an understandable public story. It uses UDISE+ 2023–24 national highlights from the Ministry of Education, and points to the Open Government Data Platform India for follow-up enrolment and infrastructure datasets. The analysis includes four charts, a plain-language data story, social posts, presentation material, a quiz, and a Python chart-refresh script. AI output is treated as a draft; official source documents remain the evidence. Important limits—especially the difference between GER, attendance, and completion—are stated clearly.

2. Research summary: data storytelling and visualisation

Data storytelling combines evidence, visual form, and narrative. The data should answer a defined question; the chart should make the comparison legible; and the narrative should distinguish observed patterns from explanations. For general readers, charts need descriptive titles, clear units, accessible labels, source notes, and concise takeaways. Avoid truncated bar-chart axes that exaggerate differences, 3-D effects that distort area or volume, inconsistent category definitions, cherry-picked time windows, and colour choices that imply a judgment not present in the data.

Tufte-style guidance is used here as a general design principle: maximise useful information, reduce decorative clutter, and keep visual encoding faithful to the numbers. This report does not quote a specific Tufte book passage; the principles are paraphrased. AI can help explain datasets, draft captions, suggest chart types, and automate updates, but it can also hallucinate causes, confuse metrics, or use outdated figures. Human verification is therefore mandatory.

3. Datasets and source register

Dataset / source	Variables / use	Status and limits
UDISE+ national highlights 2023–24	Schools, teachers, enrolment, GER by stage	Values transcribed from official national highlights; see source URL be
OGD: Enrolment by location, management, school category and social category	Academic year, state/district, school management/category, social category, enrolment	Catalog record verified; raw CSV not independently downloaded in this run
OGD: UDISE+ dataset group / school directory	Facilities, school management and category; teacher characteristics, selected datasets	Catalog verified; select latest year and validate definitions befor
Census of India 2011 literacy tables	(optional context)	Older reference year; do not compare as if current or mix with UDISE-

https://www.education.gov.in/sites/upload_files/mhrd/files/statistics-new/udise_report_existing_23_24.pdf

https://www.education.gov.in/sites/upload_files/mhrd/files/statistics-new/udise_report_nep_23_24.pdf

https://www.data.gov.in/catalog/enrolment-location-school-management-school-category-and-social-category-udise-plus

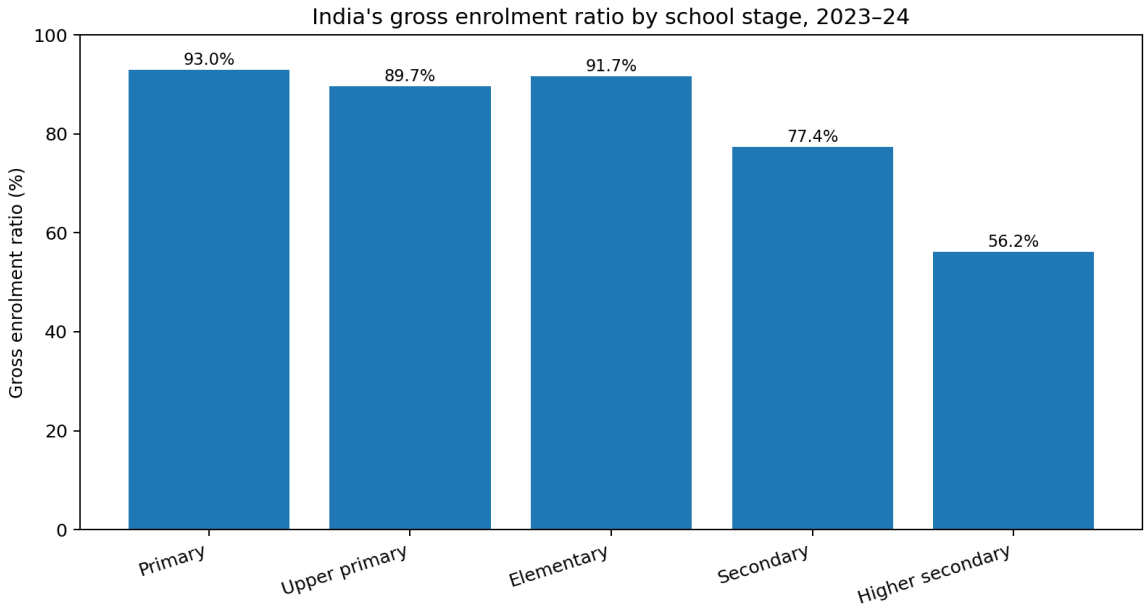
https://www.data.gov.in/dataset-group-name/Unified%20district%20information%20system%20for%20education

https://censusindia.gov.in/nada/index.php/catalog/42611/download/46274/Census%20of%20India%202011-Provisional%20Population%20Totals.pdf

4. Verified indicators used in charts

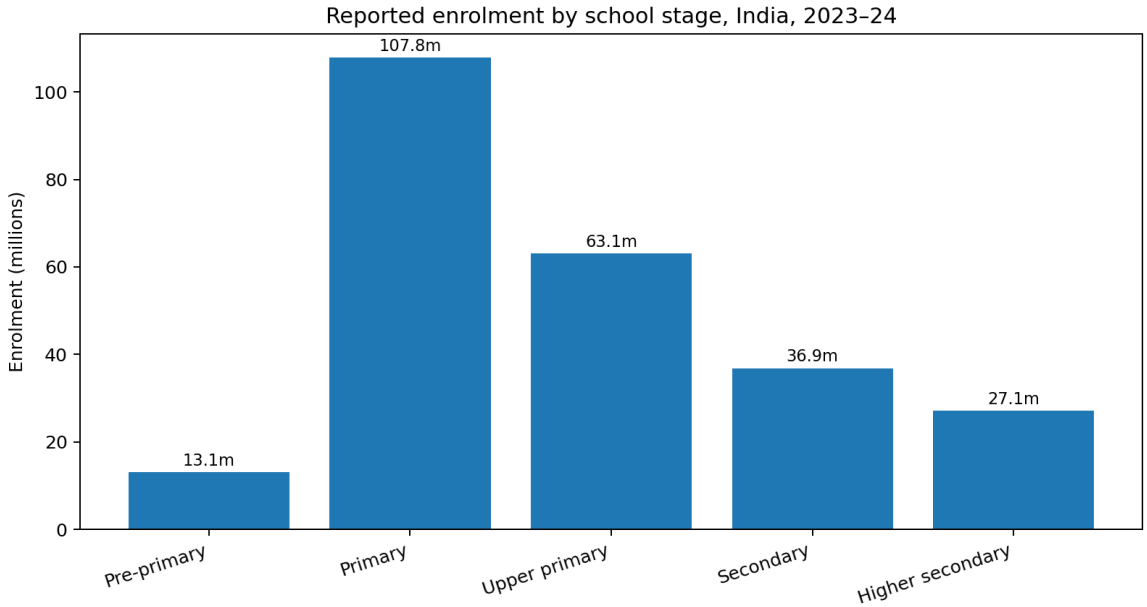
The Ministry of Education's UDISE+ 2023–24 national highlights report 1,471,891 schools, 9,807,600 teachers, 248,045,828 enrolments from pre-primary to higher secondary, and 234,963,031 enrolments from primary to higher secondary. Reported GER values are 93.0% primary, 89.7% upper primary, 91.7% elementary, 77.4% secondary, and 56.2% higher secondary. Reported enrolment from primary through higher secondary includes 113,248,084 girls and 121,714,947 boys.

The report notes that pre-primary enrolment does not include Anganwadi ECCE centres and private kindergartens outside UDISE+ coverage. GER is not an attendance or completion rate. Stage-level enrolment counts do not, by themselves, measure dropout.



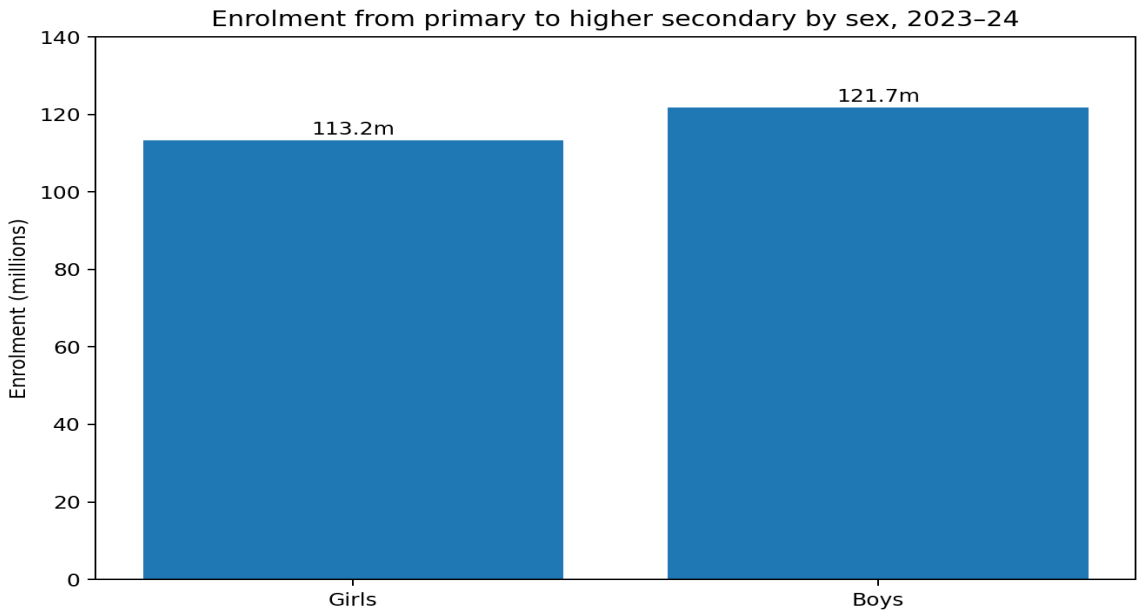
Source: Ministry of Education, UDISE+ 2023–24. GER is not the same as attendance or completion.

Chart 1. Gross enrolment ratio by school stage. Takeaway: reported GER is lower at higher-secondary than primary level; the chart does not explain why.



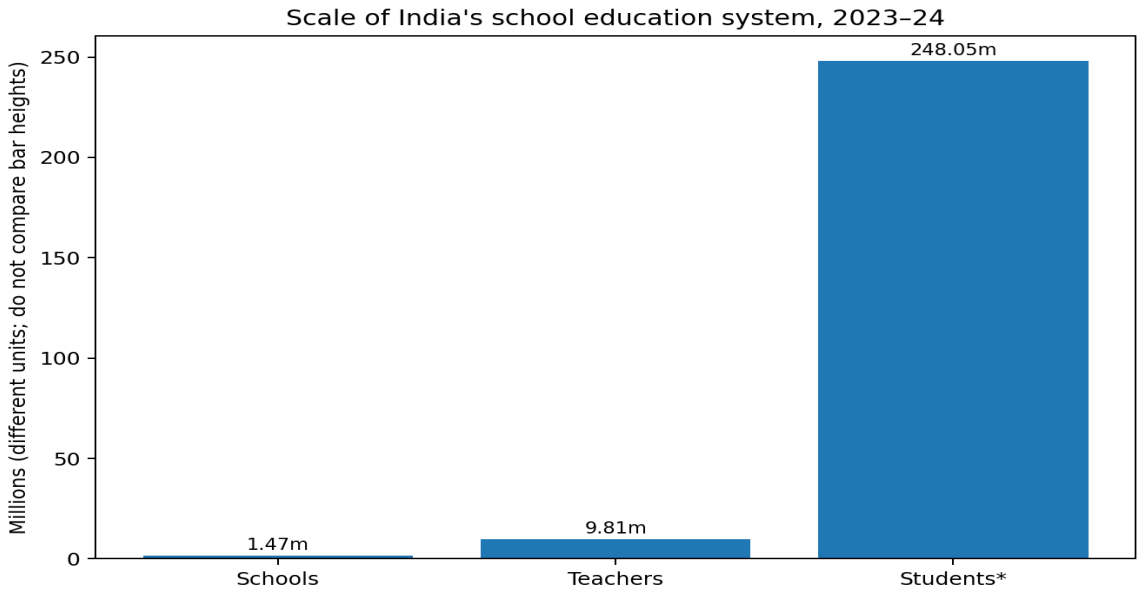
Source: Ministry of Education, UDISE+ 2023–24. Pre-primary coverage excludes some ECCE centres and kindergartens.

Chart 2. Enrolment by school stage. Takeaway: counts vary by stage, but this is not cohort tracking and must not be labelled a dropout count.



Source: Ministry of Education, UDISE+ 2023–24. These are enrolment counts, not population-adjusted rates.

Chart 3. Enrolment counts by sex, primary to higher secondary. Takeaway: counts alone are not population-adjusted measures of equal access.



*Students = pre-primary to higher secondary. Different units are shown together for scale only; bar heights are not directly compara

Chart 4. Scale snapshot. Note: schools, teachers and students are different units; do not compare their bar heights directly.

5. Data story

India’s school education system is so large that it is easy to describe it with a single impressive number. The Ministry of Education’s UDISE+ national highlights for 2023–24 report 1,471,891 schools and 9,807,600 teachers. They also report 248,045,828 enrolments from pre-primary through higher secondary. These figures show the scale of the system—but scale alone does not tell us whether every child attends regularly, learns well, or completes school.

A useful public story begins by asking what each number actually measures. Enrolment is a count of students recorded in the system. It is not the same as daily attendance, learning achievement, or successful completion. The difference matters: a chart can be factually correct and still mislead if its title encourages readers to infer more than the data support.

One of the clearest patterns in the UDISE+ 2023–24 national highlights appears in gross enrolment ratios (GER) across school stages. The reported GER is 93.0% for primary, 89.7% for upper primary, 91.7% for elementary, 77.4% for secondary, and 56.2% for higher secondary. The higher-secondary figure is lower than the primary figure, so the numbers raise a reasonable question: how can the system help students continue through the later stages of schooling?

The chart does not, by itself, explain why the values differ. GER compares enrolment at a given stage with the population of the official age group for that stage, and it can include students outside the expected age group. It is not a direct measure of attendance or completion. Therefore, it would be inaccurate to claim from these figures alone that a particular number of students drop out, or that one policy caused the difference. Those questions require additional data on transition, retention, attendance, and the population used in the denominator.

The enrolment counts add another perspective. UDISE+ reports approximately 107.84 million students in primary, 63.13 million in upper primary, 36.86 million in secondary, and 27.14 million in higher secondary. The pre-primary figure is approximately 13.08 million. These are stage-specific counts, not a tracking of the same children as they move through school. Differences between stages may reflect cohort size, age structure, school participation, stage definitions, and other factors. A simple bar chart can make the pattern visible, but should not be presented as proof that the difference between two stages is entirely caused by students leaving school.

The source itself notes a coverage limitation for pre-primary enrolment: the figure does not include pre-primary enrolment in Anganwadi early-childhood care and education centres and private kindergartens that are outside UDISE+. That limitation is not a minor footnote. If a reader compares the pre-primary total with other stages without seeing the caveat, they may interpret incomplete coverage as a complete picture of early education. Honest storytelling places this information next to the chart, not hidden in a distant appendix.

The reported enrolment from primary through higher secondary is 234,963,031. Within that total, UDISE+ reports 113,248,084 girls and 121,714,947 boys. The counts show that both girls and boys make up a very large part of the school system, while the reported boy count is higher. But counts alone do not establish equal access or unequal opportunity. To study gender equity more carefully, analysts should examine enrolment rates using appropriate population denominators, and investigate attendance, retention, completion, disability, location, household circumstances, and other relevant factors. The data should not be used to stereotype students or communities.

AI can help make these figures more accessible. A tool can draft a plain-language chart caption, generate a Python script, identify missing values, or suggest questions for further investigation. It can also make mistakes: it may confuse GER with attendance, invent a cause for a trend, or omit a source caveat. In this project, AI-generated claims are treated as drafts rather than evidence. Each statistic is checked against the Ministry of Education’s report, and each chart carries its source and reference year.

The same principle guides the visual design. Bar charts start at zero when bar length represents magnitude. Axes and units are labelled. Categories use consistent definitions. The title says what is measured rather than declaring a dramatic conclusion. The source and important limitations appear directly on the visual. Where data are missing, they are labelled as unavailable rather than silently replaced with zero. Comparisons between different years are made only after checking whether definitions and coverage are comparable.

The next step is to combine the national figures with compatible state- or district-level datasets on infrastructure, transition, and attendance. The Open Government Data Platform India lists UDISE+ datasets on enrolment and school facilities. Those datasets can help answer local questions, but only after their reference years, geographic boundaries, units, and definitions are verified. The present report does not invent state-level values or claim to have downloaded a complete current CSV when that file has not been attached and inspected.

The public lesson is simple: numbers become useful when readers can see what they mean, where they came from, and what remains unknown. India’s school system is extensive, and the 2023–24 indicators reveal differences between school stages that deserve attention. They do not provide every answer. Responsible data storytelling makes the evidence clearer without making it stronger than it is—and uses AI to support that responsibility, not replace it.

6. Five-post social media series

Post 1
■ India’s school system is vast: UDISE+ reports 1,471,891 schools in 2023–24. That number describes scale—not automatically quality, attendance, or learning. Read the data with context. Source: Ministry of Education, UDISE+ 2023–24.

Post 2
■ UDISE+ 2023–24 reports gross enrolment ratios of 93.0% at primary level and 56.2% at higher-secondary level. GER is not an attendance or completion rate, and the figures alone do not explain why the difference exists. Source: Ministry of Education.

Post 3
■ School-stage enrolment counts are not a student-tracking chart. Differences between primary and secondary totals may reflect several factors, so we should not call them ‘dropouts’ without additional evidence. Source: UDISE+ 2023–24.

Post 4
■■ UDISE+ reports 113.25 million girls and 121.71 million boys enrolled from primary to higher secondary in 2023–24. Counts are useful, but fair comparisons also need population denominators and context. Source: Ministry of Education.

Post 5
■ AI can help explain charts and automate updates—but it can also invent explanations. Verify every claim against the original dataset, show the year, cite the source, and disclose limitations. Honest data builds trust.

7. Six-slide presentation outline

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Data Storytelling with Open Government Data
Case study: School education in India
Audience: public, students, parents and local communities

Why data storytelling?
• Public datasets can be large and difficult to interpret.
• Charts make patterns easier to see, but can also mislead.
• A good story combines evidence, visual clarity and plain language.
• AI can assist—but every important claim needs verification.

Verified national snapshot (UDISE+ 2023–24)
• 1,471,891 schools
• 9,807,600 teachers
• 248,045,828 enrolments, pre-primary to higher secondary
• GER: primary 93.0%; secondary 77.4%; higher secondary 56.2%
Source: Ministry of Education, UDISE+ 2023–24.

What the charts show—and do not show
• GER is lower at higher-secondary than primary level in the reported figures.
• Stage enrolment counts are not cohort tracking.
• Enrolment counts are not attendance or learning outcomes.
• The figures alone do not explain causes or prove a dropout count.

Chart honesty and AI verification
• Label year, source, unit, population and definitions.
• Use appropriate axes; do not hide missing data.
• Separate observed patterns from causal explanations.
• Compare AI output against official tables and keep an audit trail.

Conclusion and next steps
• Public data becomes useful when readers understand both findings and limits.
• Download and verify state/district CSVs before adding local comparisons.
• Refresh charts with the included Python script.

• Cite every dataset and disclose uncertainty.
Sources: data.gov.in; Ministry of Education / UDISE+.

8. Ten-question quiz

1. What does a chart's source note help readers do?

A. Verify where the data came from

B. Prove the chart's conclusion automatically

C. Remove the need to check definitions

D. Guarantee the data are error-free

Answer: A

2. In UDISE+ 2023–24, the reported primary GER is 93.0%. What should a reader avoid concluding from this alone?

A. That the reported value is 93.0%

B. That GER is automatically the same as attendance

C. That the value belongs to a school stage

D. That the reference year is 2023–24

Answer: B

3. Why should bar charts that show magnitude generally start at zero?

A. To make the visual length proportional to the value

B. To hide small differences

C. To remove the need for units

D. To make every bar equal

Answer: A

4. The enrolment counts for primary and higher-secondary stages differ. Does that difference alone equal the number of students who dropped out?

A. Yes, always

B. No, stage counts do not track the same cohort by themselves

C. Yes, if the chart uses blue bars

D. Only if the title says “dropout”

Answer: B

5. What should an analyst do with a missing value?

A. Replace it with zero without explanation

B. Hide the row

C. Label it as missing and investigate

D. Copy a value from a different year

Answer: C

6. Which is the strongest chart title?

A. “A shocking education crisis”

B. “The government has failed”

C. “Gross enrolment ratio by school stage, India, 2023–24”

D. “The only truth about schools”

Answer: C

7. Name two details that make datasets more comparable.

Answer: Any two of: same reference period, same definitions, compatible units, matching geographic coverage, comparable denominators.

8. An AI assistant says that a lower higher-secondary GER is caused by household poverty, but the dataset contains only GER values. How should this claim be classified?

Answer: Unsupported by that dataset; additional evidence is needed to investigate causes.

9. What important limitation applies to the UDISE+ 2023–24 pre-primary enrolment figure?

Answer: It excludes enrolment in Anganwadi ECCE centres and private kindergartens outside UDISE+ coverage, as noted by the report.

10. Give one ethical practice for public data storytelling.

Answer: Examples include citing every dataset, avoiding cherry-picking, protecting privacy, disclosing limitations, avoiding stereotypes, and checking AI-generated claims.

9. Chart honesty checklist

1. Is the original source official and traceable?

2. Is the dataset title and reference year shown?

3. Are geography and population coverage clear?

4. Are units and denominators labelled?

5. Is the chart title descriptive rather than sensational?

6. Does a bar chart use a zero baseline?

7. Are axes, ticks, scales and rounding clearly labelled?

8. Are category definitions consistent across comparisons?

9. Are missing values shown as missing rather than zero?

10. Are time periods comparable in definition and coverage?

11. Does the takeaway distinguish observation from explanation?

12. Is correlation kept separate from causation?

13. Are data limitations placed near the chart?

14. Are colours legible, accessible, and not the only way to convey meaning?

15. Is the chart understandable to a non-specialist reader?

16. Has a human checked every number against the source?

10. Automation and prompt library

The included refresh_charts.py reads a local CSV, checks required columns, warns about missing values, and regenerates a GER chart. It does not fetch current web data automatically. To use it, install pandas and matplotlib, then run: python refresh_charts.py udise_2023_24_national_highlights.csv. The included prompt_library.md contains ten prompts for source discovery, data documentation, chart selection, caption drafting, story drafting, adversarial fact-checking, social

posts, automation, before/after review, and ethics.

11. AI verification report and ethics

Figures in this pack were checked against the Ministry of Education UDISE+ 2023–24 national highlights retrieved from the official government website. The data.gov.in catalogue entries were verified as source pages, but the raw state/district datasets were not downloaded and validated here. Before final submission, record the access date and page/table number, download the selected CSVs, recalculate results, and verify definitions.

Ethical rules: cite every dataset; avoid cherry-picking; disclose missing values and scope; do not infer causes from correlation; do not publish personal student-level data; avoid stigmatizing communities; use accessible charts; retain human review; and clearly identify AI-assisted drafting. Never treat generated text as a primary source.

12. References

Ministry of Education, Government of India. UDISE+ 2023–24, Existing Structure — National Highlights. https://www.education.gov.in/sites/upload_files/mhrd/files/statistics-new/udise_report_existing_23_24.pdf

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Tufte, Edward R. The Visual Display of Quantitative Information. General visual-integrity principles paraphrased; consult the edition available to the student for full bibliographic details.